Exercise Pegasus

an after action review

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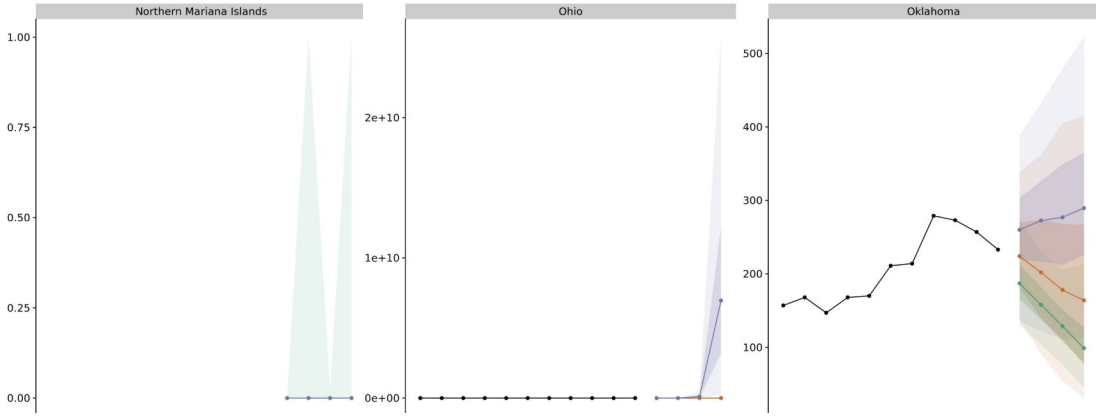
Introduction

Background: *crisis collaboration*

- Epi, dengue, COVID-19
- Multi-model collaborations (UK, US, EU)
- Model comparison < evaluation < ethics



many methods of being wrong:
COVID-19 deaths in 3 US states, Feb 2021

Research focus: *marginal value of ± modelling*

- *modelling as* evidence: model evaluation
- *modelling as* advice: synthesis and expertise
- *modelling as* a profession: ethical structures

Exercise Pegasus: an after action review

What was supposed to happen?

What actually happened?

What do we change?

Pegasus : What was supposed to happen?

UK emergency expertise

Cabinet Office > SAGE > ± experts

- SPI-M-O: ad hoc, independent, unpaid
  - COVID-19: scale, scope; 'consensus'
- "Tier 1 exercises": pandemic simulation
  - Whole-government & NHS drill
  - 2016: Ex Cygnus (H2N2)
  - Mandated by COVID Inquiry

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"…enhances the ability of UKHSA…"

- Research
  - *"focus on **collaboration…** growing UKHSA's scientific expertise"*
- Response
  - *"deliver **responsive research** [in] emergencies"*
  - Small % funded staff time

Pegasus:
UKHSA red/blue team
3 days of "play", Sep-Nov 2025

Pegasus : What actually happened?

UK emergency expertise

"ENVID-25" novel enterovirus

- SPI-M-O ~ *"best estimates of key epidemiological parameters"*
  - Unclear role vs UKHSA
  - Little coordination among tasks/methods

- Task ambiguity > Group dynamics
  - anchoring on hypotheses/estimates

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Just re-established

- Institution-led response
  - Internally fragmented; existing structures reinforced
  - ? ad hoc UKHSA interaction

- Both now developing institution-level policy for emergency response

based on only 32 case contact pairs. The serial interval estimates produced by SPI-M-O modelling groups were broadly consistent with UKHSA's estimate. The estimates produced by some modelling groups were slightly higher and lower.

Pegasus : What do we change?

UK emergency expertise

- What do we expect?
  - Pandemic simulation every 3 years
  - High political value of SPI-M as source of expertise
- What do we want?
  - Modelling as public health: UKHSA role/scope?
  - Modelling as research: better infrastructure for coordination?

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- What do we expect?
  - Given 5-year funding, will see another
  - Given aims, a greater role than academia
- What do we want?
  - Response mode: trade-off in allocating that small % time?
  - Research mode: analytics for routine vs crisis operations?

…And what's the smallest next step?

Reflections

UK emergency expertise

- SPI-M?

  "Knights of the round table"

  or "Champion/Challenger"

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- UKHSA support?

  Surge capacity

  Methodological review

  Strategy

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Appendix

The objectives for Exercise Pegasus set specific elements where there is a reasonable prospect of useful learning rather than assessing the totality of pandemic arrangements. The objectives as agreed are:

- *Exercise decision making processes for measures to contain, control or mitigate the impact of a pandemic including how relevant priority lessons from previous civil emergency exercises and outbreaks of disease have been embedded.*
- *Explore the impact of inequalities, and their consideration within pandemic decision making.*
- *Investigate likely impacts of HMG decisions on the health and care system, local responders, communities, businesses, civil society and the general public.*
- ***Examine processes for the scaling up of relevant capabilities that would be needed as part of a central government pandemic response.***
- *Investigate the effectiveness of coordination, including the exchange of information, between different tiers in a UK wide response.*
- *Test the strategic response to disinformation and misinformation.*
- *Enable and support local exercising of selected capabilities and response issues.*
- *Identify and report on relevant areas of learning to improve future pandemic preparedness.*

While the number of objectives is relatively high, it should be noted that they will not all be addressed at the same stage of the Exercise, with some to be addressed via table-top exercises in between 'live' phases or in preparatory exercises prior to the main Exercise.

Source:
https://www.england.nhs.uk/long-read/pandemic-preparedness-exercise-pegasus/

DHSC are leading overall on the Pandemic Preparedness programme, and NHS England (via NHS Resilience, programme teams and National Clinical Directors) are supporting in partnership with UKHSA, Cabinet Office, and other ALBs

- How much do we need to know about civil service operations in order to produce evidence effectively?